
Solutions Data 250 Midterm 1 9/26/25

Q1

(20) **1.** Find a linear recurrence that gives this sequence.

Out[]=

1, 1, 2, 5, 11, 24, 53, 117, 258, 569

A1

`FindLinearRecurrence[{1, 1, 2, 5, 11, 24, 53, 117, 258, 569}]`

Out[]=

{2, 0, 1}

$a[n+3] = 2 a[n+2] + a[n]$

Must check all cases.

Q2

(15) **2.** Simplify:

$(A \wedge B) \wedge \neg(\neg A \wedge B)$

A2

`Simplify[Or[And[A, B], Not[And[Not[A], B]]]`

Out[]=

$A \vee B$

$A \vee \neg B$

Q3a

3a. (10) Write a short python program to define nor and implies where the inputs to the functions are two boolean variables.

A3a

```
def nor(x, y):  
    return(not(x or y))
```

```
def implies(x,y):  
    if(x):  
        return(y)  
    return(True)
```

Q3b

b. (10) Then write a short python program to show the truth tables of these two expressions are the same.

$((\text{not } p) \text{ nor } (\text{not } q)) \text{ nor } (p \text{ nor } q)$

$(p \text{ implies } q) \text{ implies } ((\text{not } p) \text{ implies } (\text{not } q))$

A3b

```
result = True
booles=[True, False]
for p in booles:
    for q in booles:
        if(xor(((not p) nor (not q)) nor (p nor q), (p implies q) implies ((not p) implies (not q)))):
            result=False
```

Q4

(15) 4. Give a closed form for

$Out[] =$

$$\sum_{k=1}^n (2k + 3)$$

A4

$\text{Sum}[2k + 3, \{k, 1, n\}]$

$Out[] =$

$$4n + n^2$$

Q5a

5. a (10) Given $x \geq 0$ write $x = 2n + r$

where $2 > r \geq 0$ and n is an integer by finding formulas for n and r in terms of x using Floor, Ceiling, Mod, and/or arithmetic.

A5a

Let $n = \lfloor x/2 \rfloor$ and $r = x \bmod 2 = x - 2n$

Then since $n \leq x/2 < n+1$

$2n \leq x < 2n+2$ we subtract $2n$ from each expression to get that $0 \leq r < 2$

Q5b

b) (5) Use the decomposition of x in part a to show that

$$\lfloor x/2 \rfloor = \lfloor \lfloor x \rfloor / 2 \rfloor \text{ for } x \geq 0$$

A5b

$\lfloor x/2 \rfloor = n$ by part (a) so we have to show the rhs is n .

The floor of x is either $2n$ or $2n+1$

Case 1: $\lfloor x \rfloor = 2n$ then $\lfloor \lfloor x \rfloor / 2 \rfloor = n$

Case 2: $\lfloor x \rfloor = 2n+1$ then $\lfloor \lfloor x \rfloor / 2 \rfloor = n + 1/2$

Since n is an integer, $\lfloor n + 1/2 \rfloor = n$

QED

Q6

(15) 6. Enumerate all numbers $0 \leq x \leq 14$ such that $x \equiv 4 \pmod{5}$

A6

```
Select[Range[0, 14], Function[x, Mod[x, 5] == 4]]
```

Out[]=

```
{4, 9, 14}
```

Extra Credit

(2) EC. Sum of the powers of 3 from 1 to 2187 (inclusive)

Answer

Use the formula for a geometric sum, $(3^{(7+1)}-1)/(3-1)$

```
Table[3^n, {n, 0, 7}]
```

Out[]=

```
{1, 3, 9, 27, 81, 243, 729, 2187}
```

```
Sum[3^n, {n, 0, 7}]
```

Out[]=

```
3280
```